

Curriculum Vitae

Dr Chinnadurai S., PhD in Biochemistry (AIIMS, New Delhi),
MBA (Hospital Management)

Consultant Biochemist at Anderson Diagnostics and Labs
Private Limited, Chennai, Tamil Nadu.

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PROFESSIONAL SNAPSHOT

- Expertise in Laboratory Quality Management System, Laboratory Diagnostic Procedures (Clinical Biochemistry and Molecular Biology), Result interpretations & Quality reporting.
- Laboratory Management which includes Regulatory compliance and safety, Data Analysis, Staff Training and Supervision, Workflow Optimization, Customer Service and Troubleshooting etc.
- Total Experience as Clinical Biochemist and additionally Quality Manager role with Managerial skill responsibilities: **> 11 years**
- Quality Assurance & Quality Control Program, participation in NABL Audits and endowed with analytical, time management, organizational and **multi-tasking skills including professional teaching, Life science research and strong communicating skills.**

Expertise in Diagnostic	: Medical Laboratory Specialist (NABL authorized signatory for Clinical Biochemistry),
Administration	: Laboratory head (Operations),
Research	: Molecular Cancer Biology,
Teaching	: Teaching and Coordinating academic activities,
Additional Activity	: Member of hospital link staff committee for NABH.

- Completed International Lead Auditors Certificate Program and Laboratory Quality Management System (ONSITE) course as per **ISO 15189:2022** which was held during 17th and 18th of June 2023 at **Anderson Diagnostics and Labs Private Limited, Chennai, Tamil Nadu** Organized by THE FOUNDATION FOR QUALITY (INDIA).

- Acquired advanced Training on International medical laboratory quality management system (QMS) & MEDICAL LABORATORY auditors certificate course (As per the requirement of **ISO 15189:2012 & ISO/IEC 19011**) held during 22nd to 25th August 2021, Organized by THE FOUNDATION FOR QUALITY (INDIA).
- Quality protocols of Medical Laboratories Quality Management System and Internal Audit as per **IS / ISO 15189:2007** held during 05-08 February 2013 at Bureau of Indian Standards, Chennai, Tamil Nadu.
- Successfully completed regulatory audit of NABL as per **ISO 15189 & ISO/IEC 19011** as a Quality Manager, and also enrolled for NABL authorized signatory for Clinical Biochemistry division at **Regional Processing Laboratory, Thyrocare Technologies Limited**, New Delhi.
- Undergone training on **Basic Life Support** at **Centre for Medical Simulation**, SRM Medical College Hospital & Research Center, Chennai, Tamil Nadu.

CAREER RECITAL

POSITIONS HELD	INSTITUTION / ORGANIZATION
Consultant Biochemist May 2023 to till date	Anderson Diagnostics and Labs Private Limited , Greams Road, Chennai, Tamil Nadu.
Lab Head (Operations), Quality Manager and Clinical Biochemist January 2021 to May 2023	Regional Processing Laboratory, Thyrocare Technologies Limited (NABL, ISO and CAP Accredited), New Friends Colony, New Delhi. (Corporate Office: D 37 / 1, MIDC, Navi Mumbai).
Quality Manager cum Laboratory Supervisor July 2020 to December 2020	Central Clinical Laboratory, Adhiparasakthi Hospitals, (NABL accredited for Molecular Testing – Closed and Open system) Melmaruvathur, T N.
Biochemist cum Manager July 2015 to June 2020	Central Clinical Laboratory, SRM Medical College Hospital and Research Centre, SRM Institute of Science and Technology, Chennai, T N.
Senior Registrar September 2012 to March 2016	Clinical Biochemistry division, SRM Institutes for Medical Sciences (SIMS Hospitals), Chennai, T N.
Post-doctoral Research Associate November 2011 to August 2012	Department of Biochemistry, All India Institute of Medical Sciences, Ansari Nagar, New Delhi.

QUALIFICATION

DEGREE	INSTITUTION / ORGANIZATION
Ph.D in Biochemistry (2006 -2011)	University of Delhi South Campus (UDSC) and All India Institute of Medical Sciences (AIIMS), New Delhi.
MBA (Hospital Management) (May 2021 – May 2023)	Alagappa University, Karaikudi, Tamil Nadu.
Senior Research Fellow (2004 – 2006)	Indian Council of Medical Research (ICMR), New Delhi.
Pre-Doctoral Research Fellow (Dec 2001 – Jun 2003)	Department of Biotechnology, Sri Ramachandra Institute of Higher Education and Research, Tamil Nadu.
Trainee – Clinical Laboratory (May 2001 – Nov 2001)	Department of Biochemistry at Sri Ramachandra Institute of Higher Education and Research, Tamil Nadu.
M.Sc in Biochemistry (2000)	Bharathidasan University (Research project was carried out at the Department of Epidemiology, The Tamil Nadu Dr. M.G.R. Medical University , Chennai, Tamil Nadu).
B. Sc in Biochemistry (1998)	Bharathidasan University, Tamil Nadu.
PGDCA (1998)	@ Bureau of Data Processing Systems Pvt. Ltd, Mumbai.

MEMBERSHIP OF PROFESSIONAL ASSOCIATIONS

- Lifetime member of Association of Clinical Biochemists of India (**ACBI**)
NO:3884/LM/JOUR-
https://acbindia.org.in/members_summary.php?id=26&user_start=110
- Lifetime & Executive member of the Clinical Biochemist Association of Tamil Nadu (**CBAT**) NO: CBAT/LM-012/2017
<https://www.cbatindia.org/about> and <https://cbatindia.org/directoryOfMembers>
- American Association for Cancer Research (**AACR**), Indian Association for Cancer Research (**IACR**) and Indian Society of Cell Biology (**ISCB**)
- Ph.D. Alumni of the Department of Biochemistry - University of Delhi South Campus:
[http://biochem.du.ac.in/web/uploads/Ph.D.%20Alumni%202018%20\(14052019\).pdf](http://biochem.du.ac.in/web/uploads/Ph.D.%20Alumni%202018%20(14052019).pdf)

LABORATORY DIAGNOSTIC EXPERIENCE

Technical Expertise – Clinical Chemistry & other diagnostic areas:-

- Pre-examination, examination, and post-examination processes for moderate and high complexity testing for non-waived tests (ISO 15189: 2012). Reviewing all documentation process for daily, weekly and monthly QC and records on maintenance of instruments to ensure accuracy and precision as per guidelines.
- Complete operation and result interpretation of the following analytic methods - Photometry, Immuno Assay methods (CLIA, CMIA & ECLIA), ELISA and Immunofluorescence assay (IFA), Western Blotting, DNA, RNA, Protein isolation, purification and extraction, RT -PCR, Flow cytometry & Hemogram analysis, Capillary Electrophoresis, HPLCs, Nephelometry, LC-MS, ICP-MS.
- Clinical Biochemistry fully auto analyzers - Beckman Coulter Olympus AU 5800, Siemens Advia 2400 and 1800, Roche COBAS C311, BIOSYSTEMS.
- Chemiluminescent Immunoassays (CLIA) - (Siemens Advia Centaur XP and XPT, Immulite 2000 XPi, Abbott Architect i2000, Roche EEE Modular, COBAS E411 and e801 and VITROS), Fluorescent Immunoassays (Biomérieux, MiniVIDAS).
- Protein microarray technology of FOOD Intolerance - Quantitative assay for investigation of IgG mediated food sensitivity by Cambridge Nutritional Sciences (CNS).
- High Performance Liquid Chromatography (HPLC): Bio-Rad VARIANT II TURBO & Flow Cytometry / Fluorescence Flow Cytometry (BD FACS Calibur & SYSMEX XN - 1000)
- LC MS / MS - 8040 & 8045 from Shimadzu with a Quadrupole MS detector.
- Inductively Coupled Plasma - Mass Spectrometry (ICP - MS) iCAP Q ICP-MS (Thermo Fisher Scientific Inc.) & Agilent 7700 (Agilent Technologies).
- Antenatal analysis ANA17B - Immunofluorescence Technology - HELMED Analyzer and AESKU Scan.

Technical Expertise – Molecular Biology

- Molecular testing by real time RT-PCR – Open system (Quant Studio, BIORAD CFX 96 & c1000 Touch Thermocycler) and closed system (TRUENAT TRUELAB QUATTRO from MOLBIO). RNA extraction - Automated QIA Symphony and Kingfisher.
- COVID-19 testing and HCV RNA RT-PCR (Qualitative and Quantitative) – NABL accredited and ICMR approved.

RESEARCH EXPERIENCE

Ph.D. Thesis: 2006 – 2011 – PhD in Biochemistry under the supervision of **Prof. Subrata Sinha**, Dean (Research) & Head, Department of Biochemistry, All India Institute of Medical Sciences, New Delhi & **Prof. Debi P Sarkar**, University of Delhi South Campus, New Delhi.

Major Objectives of thesis:

Modulation of curcumin responsive pathways by hypoxia: a study in glial cells and “Influence of hypoxia on the nature of cell response to hydroxylamine either singly or in combination with cisplatin”.

Research work involves an investigation of the role of curcumin in the potentiating of cytotoxicity on human glioblastoma cell lines (U87MG, U373MG cells having wild type and mutated p53 respectively) as well as with SaOS-2 (null p53) Osteosarcoma cell line under different conditions of cell density, oxygen and carbon dioxide concentration to mimic in vivo conditions. I have also worked on to elucidate the role of hypoxia in influencing the response to an ROS enhancing agent hydroxylamine. We had seen the nature of cell response to hydroxylamine alone & in combination with cisplatin.

Additionally, worked on the project entitled “Identification and Functional Characterization of a Novel Unspliced Transcript Variant of HIC-1 in Human Cancer Cells Exposed to Adverse Growth Conditions” which has been published in **Cancer Res 2006; 66: 10466-10477**

AWARDS & EXTRACURRICULAR SKILLS

Received **G.P. Talwar Award** for best paper published in the journal **American Association for Cancer Research (AACR) 2006; 66: 10466-10477**, AIIMS, New Delhi.

- Elected to the post of **Sports Secretary** for the **Society of Young Scientists (SYS)** panel 2010 - 2011 at AIIMS, New Delhi.
- Awarded the prizes in the **intermediate badminton competition** in the national pulse program in the year 2007 and 2008 at AIIMS, New Delhi.
- **Volunteer Blood Donor** and Involved in many vaccination programs.
- **Type writing** (30 words per minute).
- **Council for Six Sigma certification (CSSC)** - Green belt certified.

RESEARCH INTERESTS

- Role of hypoxic region in cancer initiation, progression and metastases.

- Identification of genetic and epigenetic changes in cancer.
- Interested in learning more advanced techniques for prognosis of different cancers and to identify novel remedies for cancer.

PUBLICATIONS

- Vinupritha P, Hariharan M, Kathirvelu D, **Chinnadurai S**. Estimation of hemoglobin A1c using the complete blood count measures in the diagnosis of diabetes. **Asian Journal of Pharmaceutical and Clinical Research**, Vol 10 Issue 9 September 2017.
- Mondal AM, **Chinnadurai S**, Datta K, Chauhan SS, Sinha S, Chattopadhyay P. Identification and Functional Characterization of a Novel Unspliced Transcript Variant of HIC-1 in Human Cancer Cells Exposed to Adverse Growth Conditions. **Cancer Res** 2006; **66**: 10466-10477.
- **Chinnadurai S**, Sarkar DP, Chattopadhyay P, Sinha S. Cell density and Oxygen concentration may modulate cell response to curcumin an in vitro study (Manuscript in preparation).
- **Chinnadurai S**, Sarkar DP, Chattopadhyay P, Sinha S. Hydroxylamine fails to produce reactive oxygen species under hypoxic condition in human glioblastoma cell line (Manuscript in preparation).

POSTER PRESENTATION

- Mondal AM, **Chinnadurai S**, Chauhan SS, Sinha S, Chattopadhyay P. Homeostasis of HIC-1 expression and its modulation by p53 in human cancer cells exposed to adverse cellular growth conditions at 30th All India Cell Biology Conference & Symposium on “Molecules to Compartments: Cross-Talks and Networks” Feb 2007 at University of Delhi, New Delhi, India.
- Mondal AM, **Chinnadurai S**, Chauhan SS, Sinha S, Chattopadhyay P. Modulation of HIC-1 Expression By p53 Under Serum Starvation and Cisplatin Exposure, In Glioma Cell Lines. International Symposium on Translational Research: Apoptosis & Cancer Dec 2005 at Trivandrum, Kerala, India.

EXPERTISE IN RESEARCH TECHNIQUES

- Tissue culture and maintenance of cell lines (routine as well as under hypoxia).
- Flow Cytometry - DNA content by PI staining, Cell signaling procedures, Cell proliferation assays (MTT and XTT), ELISA, Plasmid and genomic DNA isolation, Primer designing, PCR, Real Time PCR, RNA extraction.

- Basic techniques like Polyacrylamide gel electrophoresis, Agarose gel electrophoresis, Immunoprecipitation and Western blotting.
- Immunocytochemistry, fluorescence and Confocal / Electron Microscopy.
- Intracellular ROS detection, Dual luciferase assay and Handling of laboratory animals like Rats and Mice etc.

ORGANIZATION & LEADERSHIP EXPERIENCE

- Organizing Secretary of National Symposium “**Immunity to Infection**” held on 8th October 2007 at AIIMS, New Delhi, organized by Society of Young Scientists (SYS).
- Organizing Secretary of National Symposium “**Nanotechnology in Medicine**” held on 13th Jan 2007 at AIIMS, New Delhi, organized by Society of Young Scientists (SYS).
- Part of organizing committee of National symposium “**Symposium on Molecular aspects of ... metabolic disorders**” held on 24th October 2005 at AIIMS, New Delhi.

CONFERENCES / SYMPOSIUM ATTENDED (Selected)

- **Roche Quality Quorum QC Experts Meet – 2023** which was organized by Roche on 11th September 2023 at Courtyard, Chennai.
- International Lead Auditors Certificate Program and Laboratory Quality Management System (ONSITE) course as per **ISO 15189:2022** which was held during 17th and 18th of June 2023 at **Anderson Diagnostics and Labs Private Limited**, Chennai, Tamil Nadu Organized by THE FOUNDATION FOR QUALITY (INDIA).
- International medical laboratory quality management system (QMS) & MEDICAL LABORATORY auditors certificate course (As per the requirement of **ISO 15189:2012 & ISO/IEC 19011**) held during 22nd to 25th August 2021, Organized by THE FOUNDATION FOR QUALITY (INDIA).
- Bioethicon – 2019 held on 8th to 10th Nov 2019 at SRM Medical College Hospital and Research Centre, SRM Institute of Science and Technology.
- Quality control practice in clinical laboratory held on 6th Aug 2018 at Department of Biochemistry, SRM Medical College Hospital and Research Centre, SRM University.
- Participated in “Total Quality Management in Lab medicine” – CME conducted by the Department of Biochemistry and Hematology, The Madras Medical Mission, Chennai on 31.01.2015. CME credits accredited by the Tamil Nadu Dr.M.G.R Medical University, Chennai – 5 points.
- CME on Pediatric Liver Diseases held at SIMS hospitals (SRM Institutes for Medical Science) on 28th November 2014 conducted by IMA – SIMS, Vadapalani, Chennai, TN.

- CME program and “Get – together on Molecular Biology” organized by Department of Biochemistry, SRM Medical College and Research Centre, SRM University and The Association of Medical Biochemists of India, held on April 30th 2014.
- CME program and “Get – together on Lipoproteins” organized by Department of Biochemistry, SRM Medical College and Research Centre, SRM University held on April 6th 2013.
- An American Association for Cancer Res International Conference on “New Horizons in Cancer Res: Biology to Prevention to Therapy” held at ND on Dec 13-16, 2011.
- Third National Bioethics Conference (NBC 3) held at AIIMS, ND on 17-20th Nov, 2010.
- MEDICINE UPDATE 2010 held at AIIMS, New Delhi on 8th August, 2010. This conference is accredited for 5.15 hours by Delhi Medical Council.
- SYS LECTURE SERIES 2010 entitled “The role of small RNA mediated regulatory mechanisms in tumor growth” held on 28th June 2010 at AIIMS, New Delhi.
- National symposium on TRANSLATIONAL RESEARCH IN HEALTH SCIENCES held on 24th November 2009 at AIIMS, New Delhi.
- National symposium on IMMUNITY TO INFECTION held on 8th October 2007 at AIIMS, New Delhi.
- 7th INDO-US FLOW CYTOMETRY WORKSHOP Proliferation, Apoptosis and Signal Transduction held from 10th to 14th February 2007 at School of Biotechnology, Jawaharlal Nehru University, New Delhi.
- 30th All India Cell Biology Conference & Symposium on “Molecules to Compartments: Cross-Talks and Networks” held from 2nd to 4th February 2007 at Dept of Zoology, University of Delhi, Delhi-110 007.
- 33rd Indian Immunology Society Conference held from 28th to 31st January 2007 at AIIMS, New Delhi.
- International Symposium on Translational Research: Apoptosis & Cancer held from 18th to 21st December 2005 at Trivandrum, Kerala.
- CME on Stem Cell Therapy held on 26th, 27th November 2005, AIIMS, New Delhi.
- Symposium on Molecular Aspects of Atherosclerosis and Related Metabolic Disorders held on 24th October 2005 at AIIMS, New Delhi.
- 24th Annual Convention of “Indian Association for Cancer Research” and International Symposium on “Human Papillomavirus and Cervical Cancer” held from February 9-12, 2005 at Institute of Cytology and Preventive Oncology (ICMR), Noida.

- World Federation of Neurological Surgeons (WFNS) Tumor Section Meeting-2004 held from October 11-13, 2004 at Jaipur, India.

PERSONAL DETAILS

Marital Status : Married

Nationality : Indian

Permanent Address : S/O RM. Sivasamy,
MIG – 283, NH – 1, Kulasekara alwar Street,
Maraimalai Nagar, Chengalpattu (DT),
Chennai, Tamil Nadu, India – 603209.

Passport details : NO – Y1879260
Place of Birth - Pudukkottai,
Place of issue – Tiruchirappalli
Date of Issue: 10.05.2024
Date of Expiry: 09.05.2034

Languages Known : **English** (Full professional proficiency),
Tamil (Native or bilingual proficiency) and
Hindi (Fluent in speaking)

REFERENCES

Prof. Subrata Sinha M.D, Ph.D.,
Dean (Research) & Head, Dept of Biochem (Rtd),
AIIMS, Ansari Nagar, New Delhi.
(Former *Director*, NBRC,
Manesar, Gurgaon, India).
Email ID: subrata.sinha@gmail.com

Dr.T. Swaminathan MS (Gen Surgery),
Medical Superintendent,
Madha Medical College and Hospital,
Kundrathur Main Road, Kovur,
Tamil Nadu, India – 600 122.
Email ID: gomugi@rediffmail.com

Prof. Debi P Sarkar Ph.D.,
Head, Dept of Biochemistry (Rtd),
University of Delhi South Campus (UDSC),
New Delhi, India – 110021.
(Former *Director of IISER*, Mohali),
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Dr.Vasantha Muthuswamy M.D.,
Senior Deputy Director General (Rtd) &
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